

COORDINATE SYSTEMS (L2)

Cartesian (x, y, z) , $x, y, z \in (-\infty, \infty)$

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$$
$$d\ell = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$$

Cylindrical (ρ, ϕ, z) , $\rho \geq 0, 0 \leq \phi < 2\pi$

$$\mathbf{A} = A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$$
$$x = \rho \cos \phi, \quad y = \rho \sin \phi$$
$$d\ell = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z$$
$$dv = \rho d\rho d\phi dz$$

Spherical (r, θ, ϕ) , $r \geq 0, 0 \leq \theta \leq \pi, 0 \leq \phi < 2\pi$

$$\mathbf{A} = A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi$$
$$d\ell = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$
$$dv = r^2 \sin \theta dr d\theta d\phi$$

VECTOR ALGEBRA (L2)

Dot Product

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta = A_x B_x + A_y B_y + A_z B_z$$

Cross Product

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \sin \theta \hat{n} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Right-hand rule: $\mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z, \mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x, \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$

DEL OPERATOR (L4)

Gradient (scalar $\rightarrow$ vector): $\mathbf{E} = -\nabla V$

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

Divergence (vector $\rightarrow$ scalar)

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Curl (vector $\rightarrow$ vector)

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Laplacian & Poisson's Eq.

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_v}{\epsilon_0}$$

Electrostatic Potential

$$\mathbf{E} = -\nabla V, \quad V = -\int_a^b \mathbf{E} \cdot d\ell$$

Divergence Theorem

$$\oint_S \mathbf{A} \cdot d\mathbf{s} = \iiint_V (\nabla \cdot \mathbf{A}) dv$$

Stokes' Theorem

$$\oint_C \mathbf{A} \cdot d\ell = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s}$$

MAXWELL'S EQUATIONS – INTEGRAL FORM (L3)

Law	Integral Form
Gauss (E)	$\oint \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enc}} = \iiint \rho_v dv$
Gauss (B)	$\oint \mathbf{B} \cdot d\mathbf{s} = 0$
Faraday	$\oint \mathbf{E} \cdot d\ell = -\frac{d}{dt} \iint \mathbf{B} \cdot d\mathbf{s}$
Ampère	$\oint \mathbf{H} \cdot d\ell = I_{\text{enc}} + \frac{d}{dt} \iint \mathbf{D} \cdot d\mathbf{s}$

MAXWELL'S EQUATIONS – DIFFERENTIAL FORM (L4)

Free Space	In Materials
$\nabla \cdot \mathbf{E} = \rho_v / \epsilon_0$	$\nabla \cdot \mathbf{D} = \rho_v$
$\nabla \cdot \mathbf{B} = 0$	$\nabla \cdot \mathbf{B} = 0$
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$

CONSTANTS & FIELD UNITS

$\epsilon_0 = 8.85 \times 10^{-12}$ F/m	$\mu_0 = 4\pi \times 10^{-7}$ H/m
$c = 1/\sqrt{\mu_0 \epsilon_0} \approx 3 \times 10^8$ m/s	$\eta_0 = \sqrt{\mu_0/\epsilon_0} \approx 377 \Omega$
$\mathbf{E}$ [V/m], $\mathbf{D}$ [C/m ²]	$\mathbf{H}$ [A/m], $\mathbf{B}$ [T]
ρ_v [C/m ³], $\mathbf{J}$ [A/m ²]	$\mathbf{S} = \mathbf{E} \times \mathbf{H}$ [W/m ²]

LORENTZ FORCE & COULOMB (L3,L5)

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r \quad (\text{point charge})$$

APPLYING GAUSS'S LAW (L5)

Steps: identify symmetry $\rightarrow$ Gaussian surface $\rightarrow \epsilon_0 \oint \mathbf{E} \cdot d\mathbf{s} = Q_{\text{enc}}$

Infinite Line Charge ρ_ℓ [C/m]

$$\mathbf{E} = \frac{\rho_\ell}{2\pi\epsilon_0 \rho} \mathbf{a}_\rho$$

Infinite Sheet ρ_s [C/m²]

$$\mathbf{E} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_n$$

Uniformly Charged Sphere (radius R , charge Q)

$$\mathbf{E} = \begin{cases} \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r & r > R \\ \frac{Qr}{4\pi\epsilon_0 R^3} \mathbf{a}_r & r < R \end{cases}$$

Inside conductor: $\mathbf{E} = 0$. $\nabla \cdot \mathbf{E} = 0 \Rightarrow$ no charge (but $\mathbf{E} \neq 0$ possible)

ELECTROMAGNETIC WAVES (L6)

Displacement Current (Maxwell's key addition)

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad [\text{A/m}^2]$$

Wave Equation (x-propagating plane wave)

$$\frac{\partial^2 E_y}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_y}{\partial t^2}, \quad \frac{\partial^2 B_z}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B_z}{\partial t^2}$$

Plane Wave Solutions

$$E_y = E_{\text{max}} \sin(kx - \omega t), \quad B_z = B_{\text{max}} \sin(kx - \omega t)$$

Wave Parameters

$$k = \frac{2\pi}{\lambda}, \quad \omega = 2\pi f, \quad f\lambda = \frac{\omega}{k} = c$$

General wave: $y(x, t) = A \cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0\right)$

TIME-DOMAIN EM WAVES (L6,L7)

z-propagating plane wave

$$\mathbf{E}(z, t) = E_{\text{max}} \cos(\omega t - \beta_0 z + \phi_0) \mathbf{a}_x \text{ [V/m]}$$
$$\mathbf{H}(z, t) = H_{\text{max}} \cos(\omega t - \beta_0 z + \phi_0) \mathbf{a}_y \text{ [A/m]}$$
$$\beta_0 = \omega/c = 2\pi/\lambda \text{ [rad/m]}; \quad \text{Wavefront: locus of equal phase}$$

PHASOR REPRESENTATION (L7)

Time-harmonic substitution ($\partial/\partial t \rightarrow j\omega$)

$$\mathbf{E}(\mathbf{r}, t) \rightarrow \hat{\mathbf{E}}(\mathbf{r})e^{j\omega t}, \quad \hat{\mathbf{E}} = E_0 e^{j\theta}, \quad \mathbf{E} = \text{Re}[\hat{\mathbf{E}}e^{j\omega t}]$$

Complex: $z = a + jb = re^{j\phi}, r = \sqrt{a^2 + b^2}, \phi = \arctan(b/a)$

Maxwell's in Phasor Domain

$$\nabla \times \hat{\mathbf{E}} = -j\omega\mu\hat{\mathbf{H}}, \quad \nabla \times \hat{\mathbf{H}} = (\sigma + j\omega\epsilon)\hat{\mathbf{E}}$$

Wave Impedance

$$\eta = \frac{|\mathbf{E}|}{|\mathbf{H}|} = \sqrt{\frac{\mu}{\epsilon}}, \quad \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 377 \Omega$$

Poynting Vector (power density)

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \text{ [W/m}^2\text{]}, \quad \langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}[\hat{\mathbf{E}} \times \hat{\mathbf{H}}^*]$$

Wave Polarization

Linear: $\hat{E}_x, \hat{E}_y$ same phase

Circular: $|\hat{E}_x| = |\hat{E}_y|$, phase diff = $\pm 90^\circ$

Elliptical: $|\hat{E}_x| \neq |\hat{E}_y|$ or phase diff $\neq 0^\circ, 90^\circ$

MAXWELL'S EQS. IN MATERIALS (L8)

Ohm's Law (Microscopic)

$$\mathbf{J} = \sigma \mathbf{E} \quad [\text{A/m}^2], \quad R = \rho \frac{L}{A} \text{ } [\Omega], \quad G = \sigma \frac{A}{L} \text{ [S]}$$

Electric Polarization & Permittivity

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0(1 + \chi_e) \mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E} = \epsilon \mathbf{E}$$
$$\epsilon_r = \epsilon/\epsilon_0 = 1 + \chi_e \text{ (relative permittivity)}$$

Polarization: electronic, ionic, dipole (orientation), space-charge

Magnetic Polarization & Permeability

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m) \mathbf{H} = \mu_0 \mu_r \mathbf{H}$$
$$\mu_r = \mu/\mu_0 = 1 + \chi_m; \quad \text{Dia: } \mu_r < 1; \text{ Para: } \mu_r > 1; \text{ Ferro: } \mu_r \gg 1$$

Constitutive Relations in Materials

$$\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}, \quad \mathbf{B} = \mu_0 \mu_r \mathbf{H}, \quad \mathbf{J} = \sigma \mathbf{E}$$

BOUNDARY CONDITIONS (L8)

Field	General	Perfect Conductor
Tang. $\mathbf{E}$	$E_{t1} = E_{t2}$	$E_t = 0$
Normal $\mathbf{D}$	$D_{n1} - D_{n2} = \rho_s$	$D_n = \rho_s$
Tang. $\mathbf{H}$	$H_{t1} - H_{t2} = J_s$	$H_t = J_s$
Normal $\mathbf{B}$	$B_{n1} = B_{n2}$	$B_n = 0$

EM WAVES IN MEDIA

Wave parameters in medium ($\epsilon_r, \mu_r, \sigma$)

$$v = \frac{c}{\sqrt{\mu_r \epsilon_r}}, \quad \eta = \eta_0 \sqrt{\frac{\mu_r}{\epsilon_r}}, \quad \tan \delta = \frac{\sigma}{\omega \epsilon}$$
$$\hat{\epsilon} = \epsilon \left(1 - j \frac{\sigma}{\omega \epsilon}\right) \quad (\text{complex permittivity})$$

SYMBOL REFERENCE

Symbol	Meaning	Units
$\mathbf{E}$	Electric field intensity	V/m
$\mathbf{D} = \epsilon \mathbf{E}$	Electric flux density	C/m ²
$\mathbf{H}$	Magnetic field intensity	A/m
$\mathbf{B} = \mu \mathbf{H}$	Magnetic flux density	T
ρ_v	Volume charge density	C/m ³
ρ_s	Surface charge density	C/m ²
$\mathbf{J}$	Conduction current density	A/m ²
$\mathbf{J}_d = \partial \mathbf{D} / \partial t$	Displacement current	A/m ²
σ	Conductivity	S/m
$\epsilon_r = \epsilon/\epsilon_0$	Relative permittivity	–
$\mu_r = \mu/\mu_0$	Relative permeability	–
χ_e, χ_m	Electric, magnetic susceptibility	–
$\eta = \sqrt{\mu/\epsilon}$	Intrinsic impedance	Ω
$k = 2\pi/\lambda$	Wave number	rad/m
$\beta_0 = \omega/c$	Phase constant (free space)	rad/m
$\omega = 2\pi f$	Angular frequency	rad/s
$\mathbf{S} = \mathbf{E} \times \mathbf{H}$	Poynting vector	W/m ²

EX 1: VECTOR OPS (L2)

Given: $\mathbf{A} = 2\mathbf{a}_x - \mathbf{a}_z$, $\mathbf{B} = 3\mathbf{a}_y$. $\mathbf{A} + \mathbf{B}$, $\mathbf{A} \cdot \mathbf{B}$, $\mathbf{A} \times \mathbf{B}$

$$\mathbf{A} + \mathbf{B} = 2\mathbf{a}_x + 3\mathbf{a}_y - \mathbf{a}_z$$

$$\mathbf{A} \cdot \mathbf{B} = (2)(0) + (0)(3) + (-1)(0) = \boxed{0}$$

(perpendicular $\Rightarrow$ zero dot product)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 2 & 0 & -1 \\ 0 & 3 & 0 \end{vmatrix} = \boxed{3\mathbf{a}_x + 6\mathbf{a}_z}$$

$$\mathbf{B} \times \mathbf{A} = -3\mathbf{a}_x - 6\mathbf{a}_z \text{ (anticommutative)}$$

EX 2: DIVERGENCE, CURL, GRADIENT (L4)

Given: $\mathbf{A} = x^2\mathbf{a}_x + xyz\mathbf{a}_y + z^2\mathbf{a}_z$, $V = x^2y + yz$

Divergence

$$\nabla \cdot \mathbf{A} = \frac{\partial(x^2)}{\partial x} + \frac{\partial(xyz)}{\partial y} + \frac{\partial(z^2)}{\partial z} = \boxed{2x + xz + 2z}$$

Curl (z-component)

$$(\nabla \times \mathbf{A})_z = \frac{\partial(xyz)}{\partial x} - \frac{\partial(x^2)}{\partial y} = yz - 0 = \boxed{yz}$$

Gradient

$$\nabla V = 2xy\mathbf{a}_x + (x^2 + z)\mathbf{a}_y + y\mathbf{a}_z$$

$$\mathbf{E} = -\nabla V = -2xy\mathbf{a}_x - (x^2 + z)\mathbf{a}_y - y\mathbf{a}_z$$

EX 3: GAUSS – LINE CHARGE (L5)

Given: $\rho_\ell = 5 \text{ nC/m}$ on z-axis. Find $\mathbf{E}$ at $\rho = 2 \text{ m}$.

Solution

Symmetry: cylindrical. Gaussian surface: cylinder (ρ , L).Flux only through curved surface ($\mathbf{E} \parallel d\mathbf{s}$):

$$\varepsilon_0 E(2\pi\rho L) = \rho_\ell L$$

$$\mathbf{E} = \frac{\rho_\ell}{2\pi\varepsilon_0\rho}\mathbf{a}_\rho = \frac{5 \times 10^{-9}}{2\pi(8.85 \times 10^{-12})(2)}\mathbf{a}_\rho$$

$$\boxed{\mathbf{E} \approx 44.9 \text{ V/m } \mathbf{a}_\rho}$$

EX 4: GAUSS – SPHERE (L5)

Given: Sphere $R = 0.1 \text{ m}$, $\rho_v = 10^{-6} \text{ C/m}^3$. Find $\mathbf{E}$ inside and outside.

Total charge

$$Q = \rho_v \cdot \frac{4}{3}\pi R^3 = 4.19 \times 10^{-9} \text{ C}$$

Outside ($r = 0.2 \text{ m}$)

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\mathbf{a}_r \approx \boxed{944 \text{ V/m } \mathbf{a}_r}$$

Inside ($r = 0.05 \text{ m}$)

$$\mathbf{E} = \frac{\rho_v r}{3\varepsilon_0}\mathbf{a}_r = \frac{10^{-6}(0.05)}{3(8.85 \times 10^{-12})}\mathbf{a}_r \approx \boxed{1886 \text{ V/m } \mathbf{a}_r}$$

Strategy: Gauss's Law

- Symmetry check
- Pick Gaussian surface
- $\mathbf{E} \parallel d\mathbf{s}$ on surface
- $\varepsilon_0 \mathbf{E} \cdot \mathbf{A} = Q_{\text{enc}}$

EX 5: EM WAVE PARAMETERS (L6)

Given: $f = 100 \text{ MHz}$ in free space. Find λ , k , ω , T .

$$\lambda = c/f = 3 \times 10^8 / 10^8 = \boxed{3 \text{ m}}$$

$$k = 2\pi/\lambda \approx \boxed{2.09 \text{ rad/m}}$$

$$\omega = 2\pi f \approx \boxed{6.28 \times 10^8 \text{ rad/s}}$$

$$T = 1/f = \boxed{10 \text{ ns}}$$

$$\text{Check: } f\lambda = 3 \times 10^8 = c \checkmark$$

EX 6: TIME-DOMAIN + FIND H (L6,L7)

Given: $\mathbf{E} = 50 \cos(10^9 t - \beta z) \mathbf{a}_x \text{ V/m}$ in free space.Find β , λ , f , $\mathbf{H}$

$$\omega = 10^9 \text{ rad/s} \Rightarrow f = \omega/2\pi \approx 159 \text{ MHz}$$

$$\beta = \omega/c = 10^9 / (3 \times 10^8) \approx \boxed{3.33 \text{ rad/m}}$$

$$\lambda = 2\pi/\beta \approx \boxed{1.885 \text{ m}}$$

 $\mathbf{E}$ in $\mathbf{a}_x$, propagates $+z \Rightarrow \mathbf{H}$ in $\mathbf{a}_y$:

$$H_{\text{max}} = E_{\text{max}}/\eta_0 = 50/377 \approx 0.133 \text{ A/m}$$

$$\boxed{\mathbf{H} = 0.133 \cos(10^9 t - 3.33z) \mathbf{a}_y \text{ A/m}}$$

EX 7: PHASOR CONVERSION (L7)

Given: $E(t) = 10 \cos(\omega t + \pi/4)$, $\omega = 2\pi \times 10^9$.Time-domain $\rightarrow$ phasor

$$\hat{E} = 10e^{j\pi/4} = 10\angle 45^\circ = 7.07 + j7.07$$

Derivative in phasor domain

$$\begin{aligned} \partial E / \partial t &\rightarrow j\omega \hat{E} = j(2\pi \times 10^9)(10e^{j\pi/4}) \\ &= 2\pi \times 10^{10} e^{j3\pi/4} \end{aligned}$$

$$\text{Time domain: } \frac{\partial E}{\partial t} = 2\pi \times 10^{10} \cos(\omega t + \frac{3\pi}{4})$$

Rule: $\partial/\partial t \rightarrow j\omega$; $\int dt \rightarrow 1/(j\omega)$

EX 8: POYNTING VECTOR (L7)

Given: $\mathbf{E} = 100 \cos(\omega t - \beta z) \mathbf{a}_x \text{ V/m}$ in free space.

Find time-averaged power density

$$\mathbf{H} = \frac{100}{377} \cos(\omega t - \beta z) \mathbf{a}_y$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} = \frac{10^4}{377} \cos^2(\omega t - \beta z) \mathbf{a}_z$$

$$\langle \cos^2 \rangle = \frac{1}{2}, \text{ so:}$$

$$\langle \mathbf{S} \rangle = \frac{E_{\text{max}}^2}{2\eta_0} \mathbf{a}_z = \frac{10^4}{2(377)} \mathbf{a}_z \approx \boxed{13.3 \text{ W/m}^2 \mathbf{a}_z}$$

EX 9: POLARIZATION (L7)

Linear: same phase

$$\hat{\mathbf{E}} = 3e^{j\pi/3}\mathbf{a}_x + 7e^{j\pi/3}\mathbf{a}_y$$

Phase diff $= 0^\circ \Rightarrow$ **Linear** $\checkmark$ Circular: equal amps, $\Delta\phi = 90^\circ$

$$\hat{\mathbf{E}} = 5e^{j\pi/2}\mathbf{a}_x + 5e^{j\pi}\mathbf{a}_y$$

$$|E_x| = |E_y| = 5, \text{ phase diff} = \pi/2 \Rightarrow \text{Circular } \checkmark$$

Elliptical: general case

$$\hat{\mathbf{E}} = 2e^{j\pi/3}\mathbf{a}_x + 5e^{j\pi/6}\mathbf{a}_y$$

$$|E_x| \neq |E_y| \text{ and } \Delta\phi \neq 0, \pi/2 \Rightarrow \text{Elliptical } \checkmark$$

EX 10: BOUNDARY CONDITIONS (L8)

Given: $\varepsilon_{r1} = 2$, $\varepsilon_{r2} = 5$, no surface charge. $\mathbf{D}_1 = 4\mathbf{a}_x + 3\mathbf{a}_z \text{ C/m}^2$, normal $\hat{n} = \mathbf{a}_z$. Find $\mathbf{D}_2$.

Solution

Normal $\mathbf{D}$ continuous ($\rho_s = 0$): $D_{n2} = D_{n1} = 3 \text{ C/m}^2$ Tangential $\mathbf{E}$ continuous:

$$E_{t1} = D_{t1}/\varepsilon_1 = 4/(2\varepsilon_0)$$

$$D_{t2} = \varepsilon_2 E_{t1} = 5\varepsilon_0 \cdot \frac{4}{2\varepsilon_0} = 10 \text{ C/m}^2$$

$$\boxed{\mathbf{D}_2 = 10\mathbf{a}_x + 3\mathbf{a}_z \text{ C/m}^2}$$

EX 11: WAVE IN MEDIUM (L8)

Given: $f = 1 \text{ GHz}$, $\varepsilon_r = 9$, $\mu_r = 1$, $\sigma \approx 0$.

$$v = c/\sqrt{\varepsilon_r} = \boxed{10^8 \text{ m/s}}; \lambda = \boxed{10 \text{ cm}}$$

$$\eta = 377/3 \approx \boxed{125.7 \Omega}; \beta \approx \boxed{62.8 \text{ rad/m}}$$

 λ compressed by $\sqrt{\varepsilon_r} = 3$ vs. free space

PROBLEM-SOLVING GUIDE

Gauss's Law

- Symmetry $\rightarrow$ Gaussian surface
- $\mathbf{E} \parallel d\mathbf{s}$; $\varepsilon_0 \mathbf{E} \cdot \mathbf{A} = Q_{\text{enc}}$

EM Wave: given $\mathbf{E}$, find $\mathbf{H}$

- Propagation dir from $\cos(\omega t - \beta z)$
- $\hat{k} = \hat{E} \times \hat{H}$ (RHR); $|H| = |E|/\eta$
- $\langle S \rangle = E_{\text{max}}^2/(2\eta) \text{ [W/m}^2\text{]}$

Phasor workflow

$$A \cos(\omega t + \phi) \rightarrow Ae^{j\phi}; \partial_t \rightarrow j\omega$$

$$\text{Back: } \text{Re}[\hat{A}e^{j\omega t}]$$

BCs at interface (normal $= \hat{n}_{12}$)

$$E_{t1} = E_{t2}; D_{n1} - D_{n2} = \rho_s$$

$$H_{t1} - H_{t2} = J_s; B_{n1} = B_{n2}$$

$$\text{PEC: } E_t = 0, B_n = 0$$

Displacement current (capacitor)

$$J_d = \varepsilon_0 \varepsilon_r \partial E / \partial t; I_d = J_d A = I_{\text{cond}}$$

Material trends

$$\varepsilon_r \uparrow: v \downarrow, \lambda \downarrow, \eta \downarrow; \mu_r \uparrow: \eta \uparrow$$

$$\sigma \uparrow: \text{lossy } (\tan \delta = \sigma/\omega\varepsilon \uparrow)$$